

with what the CAPM tells us. But Figure 10.10, Panel B, shows that it is hard to predict future betas. Past betas do a lousy job at predicting future betas. There is large variation in betas, and there is substantial sampling error.²⁷

Studies that estimate betas from other information tend to find positive risk relations. Buss and Vilkov (2012) estimate betas from options and find them to be better predictors of future betas than betas estimated from past returns. Their betas estimated from option-implied information yield a positive risk–return relation. Cosemans et al. (2012) use valuation information from accounting balance sheets to compute betas along with past returns. They also estimate a positive relation between betas and returns. Thus, the real mystery in the low-beta anomaly is actually not so much that beta does not work; it is that we have such difficulty in predicting future betas, especially with past betas.

4.4 RISK ANOMALY FACTORS

It is a straightforward extension to use these portfolio results to create a benchmark factor for the risk anomaly.

Betting against Beta

Frazzini and Pedersen (2010) construct a betting-against-beta (*BAB*) factor that goes long low-beta stocks and short high-beta stocks. Constructing a factor to trade the beta anomaly cannot be done just by taking differences of the portfolios in Figure 10.10. Remember, the differences in average returns across the beta quintiles are tiny—what’s large are the differences in Sharpe ratios across betas. Frazzini and Pedersen form their *BAB* factor by scaling the low- and high-beta portfolios by their betas:

$$BAB_{t+1} = \frac{r_{L,t+1} - r_f}{\beta_{L,t}} - \frac{r_{H,t+1} - r_f}{\beta_{H,t}}, \quad (10.17)$$

where $r_{L,t+1}$ is the return of the low-beta portfolio and $r_{H,t+1}$ is the return of the high-beta portfolio. The betas of the low-beta and high-beta portfolio at the beginning of the period (the pre-ranking betas) are given by $\beta_{L,t}$ and $\beta_{H,t}$, respectively.

Figure 10.12 shows what is going on. The horizontal line labeled “Data” is the empirical pattern of flat average returns with lagged betas in contrast to the upward-sloping line that is predicted by the “Standard CAPM.” The long position in the low-beta portfolio is levered. It takes the position where it hits the

²⁷ Blume (1975) was one of the first to document this. For formal statistics for calculating the paths of time-varying alphas and betas and their standard errors, see Ang and Kristensen (2012).